

Multi-Modal Enhanced Graph Transfer Learning for Digital Finance Fraud Detection

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Abstract

Fraudulent activities on blockchain networks threaten the integrity and reliability of decentralized finance ecosystems. Accurately identifying malicious nodes such as phishing or ransomware addresses, within large-scale blockchain transaction graphs remains a critical challenge due to their dynamic, sparse, and continuously evolving topologies. Transfer learning offers a powerful paradigm for fraud detection because many fraudulent schemes, including ransomware and phishing, are often orchestrated by overlapping actor groups that share behavioral and structural patterns across networks. Leveraging these shared representations enables knowledge transfer from previously observed fraud types to emerging ones. However, the complex and multi-modal nature of digital financial systems introduces substantial challenges for graph-based transfer learning. Fraudulent activities are shaped by diverse modalities including graph structure, transaction sequences, temporal price dynamics, and textual metadata, while distributional shifts frequently occur across time and platforms. Existing graph transfer learning methods struggle to model such multi-modal dependencies and to align divergent feature distributions. To tackle these challenges, we develop a Multi-modal Enhanced Graph Transfer Learning (MOE-GTL) framework which incorporates graph, temporal, and textual modalities for fraudulent node detection. We further introduce Temporal-aware Maximum Mean Discrepancy (TMMD), a regularization mechanism that explicitly aligns multi-modal feature distributions between source and target graphs over time. Extensive experiments reveal that our MOE-GTL model notably improves the accuracy of fraudulent node classifications on Ethereum and Solana transaction graphs. Our codes are publicly available at <https://github.com/yl3564/MOE-GTL>.

CCS Concepts

• Computing methodologies → Multi-modal learning.



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Keywords

Multi-modal graph learning, blockchains, graph transfer learning

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1 Introduction

The rapid adoption of blockchains such as Ethereum and Solana has disrupted the digital finance landscape. Their unique infrastructures support a wide range of financial activities, including decentralized exchanges, automated market-making, and digital asset transfers. Despite this, the pseudonymous and immutable nature of blockchain transactions leaves users exposed to various digital frauds [5], which undermine overall trust in decentralized finance (DeFi) [13]. In this setting, fraud detection is challenging due to the evolving structures of blockchain graphs [1, 14, 18] and limited labeled data. Existing studies show satisfactory detection, but struggle to generalize when existing and new frauds lead to varying and unique structural and behavioral characteristics vary [2, 4, 6, 8] such as Ethereum phishing and Solana rug-pull frauds. This highlights the need for approaches capable of accommodating heterogeneous graph structures and integrating information beyond raw transactional features. Recent progress in graph transfer learning and multi-modal representation learning has opened new opportunities for improving fraud detection performance in blockchain networks [4, 6]. Nevertheless, existing transfer learning frameworks often overlook the multi-modal nature of digital finance ecosystems, which include temporal price movements, evolving liquidity flows, and contextual signals embedded in blockchain-related news. Moreover, discrepancies in feature distributions between source and target graphs remain a major obstacle, especially when fraudulent behavior manifests differently across platforms. Addressing these gaps requires methods that can jointly model graph structure, temporal dynamics, and textual context while aligning heterogeneous node representations across networks. In this paper,

we propose a Multi-mOdal Enhanced Graph Transfer Learning (MOE-GTL) framework that integrates structural, temporal, and textual modalities to improve cross-platform fraud detection in Ethereum and Solana transaction graphs. Our approach introduces a TMMD module to mitigate cross-domain distribution shifts and enhance knowledge transfer between heterogeneous blockchain environments. Using extensive experiments on phishing datasets from Ethereum and rug-pull datasets from Solana, we demonstrate that MOE-GTL significantly improves cross-graph classification accuracy compared with existing transfer learning baselines. These findings underscore the value of multi-modal fusion and distribution alignment for advancing fraud detection in dynamic digital finance systems.

2 Related Works

Graph Transfer Learning. In graph transfer learning, a model trained on a source graph is adapted for use on a target graph. By transferring knowledge across domains or tasks, this paradigm significantly reduces training cost and improves downstream task performance, and has been widely studied in advancing graph representation learning. Early works [9] typically focused on multi-task learning settings, using source-domain information to develop graph representations transferable to related tasks in target domains. Beyond single-graph transfer, these studies [10] further investigate multi-network learning, aiming to derive node representations that are shared or transferable across multiple graphs or models. The recent focus has been transformed into the domain adaptation. Con-tuning [7] fine-tune the pretrained GNNs by generating graph-level templates for each class and incorporating them into a supervised contrastive learning objective. However, these approaches lack integration with multi-modal information and overlook multi-modal interactions. Consequently, there is a pressing need to develop efficient methods for extracting and understanding diverse multi-modal information, thereby enhancing their capabilities in transfer learning.

Digital Financial Fraud Detection. Blockchains have increasingly been modeled and as address graphs (comprising addresses and transactions) [1], but their decentralization and evolving structures mean fraud detection is not trivial. Recently, the introduction of non-fungible tokens (NFTs) and DeFi, has led to fraud detection being widely studied in the Ethereum blockchain using graph neural network (GNN) models. On Ethereum, [12] proposes a graph embedding algorithm, GTN2Vec, that accounts for both behavioral patterns of money launderers and structural information from the transaction network. In contrast, few studies have extended beyond Ethereum to other blockchains such as Solana, which may be attributed to the limited datasets available. [16] introduces a customized transformer for anomaly detection called BlockScan and shows that it can detect anomalous Solana transactions with high accuracy. However, most studies of Ethereum and Solana at most account for transactional and graph features, while few works incorporate multi-modal information beyond the address graphs.

3 Preliminaries

Consider a graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}\}$, where $\mathcal{V} = \{u_1, u_2, \dots, u_N\}$ is the set of N nodes and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is an edge set. Let $N \times N$ -matrix A

with entries $a_{i,j} \neq 0$ for any $e_{i,j} \in \mathcal{E}$ and $a_{i,j} = 0$, otherwise, be an adjacency matrix of $\mathcal{G}$. $X \in \mathbb{R}^{N \times d}$ denotes node features where d is the number of features (in our study, each node is associated with a 4-dimensional feature vector (i.e., $d = 4$), including transaction frequency as buyer and seller, and transaction amount as buyer and seller), and x_i denotes features of node i . $Y = [y_1, \dots, y_N]$ is a one-hot embedding matrix denoting class labels. In blockchain transaction graph data, each blockchain address is labeled as either phishing or normal. In this paper, our main objective is to develop a graph transfer learning model for the *month-to-month* node classification from a source graph $\mathcal{G}^s$ to a target graph $\mathcal{G}^t$.

4 Methodology

We propose a multi-modal enhanced graph transfer learning framework that integrates multi-modal modeling with transaction graphs, participant-wise time-series data, and financial news and temporal-aware maximum mean discrepancy (TMMD)-based approach for node (i.e., participant) representation learning. Specifically, our framework is composed of four core components: (i) Fine-grained Temporal Encoder (FTE), (ii) News-based Global-aware Encoder (NGE), (iii) Multi-modal Fusion Network (MMFN), and (iv) TMMD.

4.1 Fine-grained Temporal Encoder

The behavior of participants (whether acting as buyers or sellers) is strongly influenced by the historical price dynamics of blockchains, which serve as the foundation for their transactional decisions and network interactions. However, most existing models that characterize participant behavior primarily rely on transaction frequency and monetary volume, which overlook the underlying market dynamics and temporal dependencies. To address this, we introduce the fine-grained temporal encoder (FTE), which facilitates comprehensive node-level temporal information capture. First, for each node (either a buyer or seller) u , based on i -th transaction (either purchase or sell; where $i \in [1, \dots, I_u]$ and I_u denotes the total number of transactions of node u), we extract the corresponding past ω daily closed price information, denoted as ω_u^i . Then we can obtain the temporal sequence set for node u , i.e., $\mathcal{T}_u = [\omega_u^1, \dots, \omega_u^{I_u}] \in \mathbb{R}^{I_u \times \omega}$. However, the number of transactions varies across participants, to ensure temporal input information alignment across all nodes, we introduce a Multiple Fine-grained Temporal Adaptive Layer (MFTAL) for both data dimension shrinkage and jointly modeling multiple fine-grained temporal features, which is formulated as $\tilde{\mathcal{T}}_u = \text{MFTAL}(\mathcal{T}_u) = W\mathcal{T}_u$, where W denotes a trainable parameter and the output $\tilde{\mathcal{T}}_u \in \mathbb{R}^\omega$. Then we divide the obtained temporal input $\tilde{\mathcal{T}} = [\tilde{\mathcal{T}}_1, \dots, \tilde{\mathcal{T}}_N] \in \mathbb{R}^{N \times \omega}$ into overlapping patches of length ℓ with stride s . Each patch is linearly projected into a d -dimensional latent space, and positional embeddings are added to preserve temporal order. The final temporal embedding is formalized as follows:

$$Z_{\text{temporal}} = \text{Attn}(\text{PositionEncoding}(\text{Patching}(\tilde{\mathcal{T}}))), \quad (1)$$

where $\text{Attn}(\cdot)$ is a multi-head self-attention mechanism and $Z_{\text{temporal}} \in \mathbb{R}^{N \times n_p \times d}$ where $n_p = (\omega - \ell)/s + 1$ denotes the number of patches.

4.2 News-based Global-aware Encoder

To complement the transaction records, we extract blockchain-related news data aggregated by Google News, utilizing the GNews

Python package, to ensure that we use only reputable sources. We implement a script taking specified search terms and a time range, and then returns the date and time of publishing, article headline, and full article text for each matching article. News corresponding to the blockchain for its respective sample periods is extracted, and then grouped by calendar month. To effectively capture domain-specific context information, for each month, with financial news X_{news} , the process of obtaining the embedding is written as $H_{news} = \text{MLP}(\text{LM}(X_{news}))$, where $\text{LM}(\cdot)$ denotes a language model, $\text{MLP}(\cdot)$ denotes a multi-layer perceptron.

4.3 Multi-modal Fusion Network

Given a graph $\mathcal{G} = (X, A)$, we first adopt a GNN to get its corresponding graph node embeddings $H_{\mathcal{G}} = [h_1^{\mathcal{G}}, \dots, h_N^{\mathcal{G}}] = \text{GNN}(A, X) \in \mathbb{R}^{N \times d_{\mathcal{G}}}$. To fuse the learned transaction graph structural, fine-grained temporal, and market-level information, we propose a multi-modal fusion network which consists of two parts: (i) a graph-temporal multi-head attention (GT-MHA) module to preserve node-level embedding and fine-grained details of temporal embedding into merged embedding, and (ii) a gated fusion mechanism based on the financial news embedding which exploits complementary information between graph structural and temporal information. The process of GT-MHA can be formulated as follows.

$$\text{GT-MHA}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_K) W^O, \quad (2)$$

where $\text{head}_k = \text{Softmax}\left(\frac{Q_k K_k^T}{\sqrt{d_k}}\right) V_k W_k^V$, $Q_k = H_{\mathcal{G}} W_k^Q$ is the query matrix which derives from node embedding, $K_k = Z_{temporal} W_k^K$ and $V_k = Z_{temporal} W_k^V$ are the key, and value matrices, respectively (both are projected temporal embedding), W^O is learnable projection matrix, $\sqrt{d_k}$ is the dimensionality of the keys, and K is the number of attention heads. Then, the GT-MHA and residual connection are applied to progressively refine graph structural and temporal representations:

$$\tilde{H}_G = \text{LayerNorm}(\text{GT-MHA}(Q, K, V) + H_{\mathcal{G}}). \quad (3)$$

The gated fusion mechanism utilizes the node and temporal embeddings, and uses another linear mapping and softmax to map the financial news embedding to $g \in \mathbb{R}^2$ as $g = \text{Softmax}([\tilde{H}_G, \tilde{Z}_{temporal}] W_g + b_g)$, where $\tilde{Z}_{temporal} = [Z_{temporal}, H_{news}]$, W_g and b_g are learnable parameters. The first dimension g_1 of g represents the importance of transaction graph structural features, and the second dimension g_2 represents the importance of fine-grained temporal features ($g_1 + g_2 = 1$). Finally, the multi-modal representation H_{out} is obtained through calculating the weighted sum $\tilde{H}_G$ and $H_{temporal}$ of with weights g , i.e., $H_{out} = g[\tilde{H}_G, Z_{temporal}]^T$. For pre-training, we use the graph contrastive learning framework which employs a self-supervised learning framework that maximize the mutual information between two augmented views of the same graph via a contrastive loss in the learned latent space. We first utilize two augmentation functions $T_1, T_2 \sim \mathcal{T}$ to generate graph views $\tilde{\mathcal{G}}_1 = (\tilde{X}_1, \tilde{A}_1) = T_1(\mathcal{G})$ and $\tilde{\mathcal{G}}_2 = (\tilde{X}_2, \tilde{A}_2) = T_2(\mathcal{G})$, where $\mathcal{T}$ is the set of graph augmentation functions (e.g., adding/removing edges). We then obtain node representations for the two graph views via the MMFN $f(\cdot)$, denoted by H_{out} and H'_{out} respectively. Given the latent representations, we optimize the parameters of the MMFN

by a pre-defined contrastive loss. For any node i in $\tilde{\mathcal{G}}_1$, we aim to score the positive pairs (i, i^+) higher compared to other negative pairs (i, i^-) . Typically, the negative samples i^- are sampled from other nodes of the augmented graph views $\tilde{\mathcal{G}}_2$ in the same batch. The commonly-used InfoNCE loss can be defined as:

$$\mathcal{L}_{cl} = - \sum_{i=1}^N \log \frac{s(h_i^{out}, h_{i^+}^{out}, \tau)}{s(h_i^{out}, h_{i^+}^{out}, \tau) + \sum_{i^- \neq i} s(h_i^{out}, h_{i^-}^{out}, \tau)}, \quad (4)$$

where $s(\cdot, \cdot)$ is the cosine similarity between node representations and τ is the temperature parameter.

4.4 Temporal-aware Node Feature Adaptation

Recent studies in transfer learning highlight that aligning feature distributions between the source and target domains is essential for effective knowledge transfer. Thus, during fine-tuning, we first adopt a projector network that maps node features from the target graph into the feature space of the source graph. This mapping is defined as $P^t = X^t W_p$ where W_p is the trainable projection matrix. The objective of training the projector is to minimize the feature distribution discrepancy between the source features and the projected target features. A widely adopted approach to quantify such distributional dissimilarity in domain adaptation is the Maximum Mean Discrepancy (MMD). However, a fundamental limitation of MMD in the time-series domain is its inability to effectively capture participant-level transaction patterns and temporal dependencies. To address this, we introduce the temporal-aware maximum mean discrepancy (TMMD). Given temporal embeddings of nodes in source and target domains, i.e., $Z_{temporal}^s = [z_1^s, \dots, z_{N_s}^s]$ and $Z_{temporal}^t = [z_1^t, \dots, z_{N_t}^t]$, we compute the temporal similarity between two nodes, reflecting participant behavior within the graph or cross-graph from transaction and temporal patterns perspectives as $\phi_{i,j} = \cos(z_i, z_j)$, where z_i and z_j are temporal embeddings of nodes i and j respectively. If both nodes i and j belong to the target domain, we denote their association as $\xi_{i,j}^{t,t} = \log(1 + \frac{1}{\phi_{i,j}})$; whereas if node i is from the target domain and node j is from the source domain, the cross-domain correspondence is denoted as $\theta_{i,j}^{t,s} = \frac{\phi_{i,j} + 1}{2}$. Then the loss function based on TMMD can be expressed as:

$$\begin{aligned} \mathcal{L}_{tmmd}^t &= \frac{1}{(N^t)^2} \sum_{i=1}^{N^t} \sum_{j=1}^{N^t} \xi_{i,j}^{t,t} k(p_i^t, p_j^t) - \\ &\quad \frac{2}{N^t N^s} \sum_{i=1}^{N^t} \sum_{j=1}^{N^s} \theta_{i,j}^{t,s} k(p_i^t, x_j^s) + \frac{1}{(N^s)^2} \sum_{i=1}^{N^s} \sum_{j=1}^{N^s} k(x_i^s, x_j^s), \end{aligned}$$

where $k(\cdot, \cdot)$ denotes the kernel function. Finally, we adopt a multitask learning framework to jointly optimize all objective functions. The overall optimization objective is formulated as $\mathcal{L}_{final}^t = \alpha_0 \mathcal{L}_{cls}^t + \alpha_1 \mathcal{L}_{cl}^t + \alpha_2 \mathcal{L}_{tmmd}^t + \alpha_3 \mathcal{L}_{str}^t$ where $\mathcal{L}_{cl}^t$ denotes the contrastive loss based on the MMFN for the graph in target domain, $\mathcal{L}_{cls}^t = -\frac{1}{N^t} \sum_i [y_i \log \tilde{y}_i + (1 - y_i) \log(1 - \tilde{y}_i)]$ denotes a classification loss, and $\tilde{y}_i$ denotes the predicted class of node i in the target domain, and $\mathcal{L}_{str}^t$ is the structure-aware regularization [15].

5 Experiments

Datasets, Experimental Setup and Baselines We consider two datasets covering transaction records from Ethereum and Solana.

Table 1: Models performance comparison (accuracy in %) on Ethereum data from 2018/01 to 2019/01.

Method	2018/01→2018/02	2018/02→2018/03	2018/03→2018/04	2018/04→2018/05	2018/05→2018/06	2018/06→2018/07	2018/07→2018/08	2018/08→2018/09	2018/09→2018/10	2018/10→2018/11	2018/11→2018/12	2018/12→2019/01
GRACE _r	81.36±0.50	90.02±0.29	83.93±0.15	74.99±0.40	73.27±0.87	77.62±0.20	84.03±0.31	76.21±1.65	71.79±3.45	77.27±3.59	67.43±1.71	68.57±3.87
AdapterGNN	71.00±2.60	83.70±1.50	73.40±1.30	70.20±0.80	69.50±0.90	68.30±0.90	73.60±2.90	60.70±4.90	66.80±2.80	62.30±8.90	41.10±5.50	40.50±10.50
GTOT-Tuning	81.80±0.43	89.23±0.29	84.06±0.09	75.27±0.52	73.83±0.63	77.10±0.48	84.07±0.23	78.10±0.70	77.86±0.91	80.45±2.11	70.29±1.71	72.86±1.21
GraphLoRA	81.10±0.40	89.10±0.46	83.91±0.29	74.48±1.04	73.30±0.18	77.09±1.07	83.94±0.69	77.93±1.49	75.71±3.24	75.91±8.29	65.14±5.11	73.31±3.91
MOE-GTL (ours)	82.25±1.38	89.42±0.38	84.14±0.62	75.92±0.35	74.37±0.90	77.95±1.28	84.39±0.53	79.31±1.49	78.21±0.80	84.09±2.27	70.86±3.73	74.76±3.61

Table 2: Models performance comparison (accuracy in %) on Solana data from 2022/10 to 2022/12.

Method	2022/10→2022/11	2022/11→2022/12	2022/10→2022/12
GRACE _r	62.80±1.10	61.55±3.56	64.22±1.08
AdapterGNN	58.20±1.90	66.12±0.25	64.61±1.37
GTOT-Tuning	71.20±1.10	64.85±0.46	65.10±0.21
GraphLoRA	61.47±0.00	65.24±0.58	65.14±0.50
MOE-GTL (ours)	65.69±2.11	66.17±0.41	66.07±0.43

Table 3: Ablation studies (accuracy in %).

Method	2018/01→2018/02	2018/05→2018/06	2018/09→2018/10
MOE-GTL (ours)	82.25±1.38	74.37±0.90	78.21±0.80
MOE-GTL w/o FTE	81.67±0.96	74.31±0.82	77.14±3.43
MOE-GTL w/o NGE	81.62±0.99	74.30±0.89	74.64±4.45
MOE-GTL w/o MMFN	81.40±1.05	73.77±1.81	77.14±2.33
MOE-GTL w/o TMMD	81.51±0.95	74.27±0.72	76.07±3.70

The Ethereum dataset comprises fraudulent transaction records provided by XBlock in graph form, covering the period of 01/01/2018 to 01/19/2019, inclusive, which contains 2,058,528 unique addresses (nodes) and 8,541,503 transactions (edges). Of the total unique addresses, 1,165 (0.06%) are labeled as fraudulent, with the remaining 2,057,363 (99.94%) labeled as non-fraudulent. The Solana dataset comprises fraudulent transaction records from the SolRPDS database by [3] derived from 3.69 billion Solana blockchain transactions, covering the period of 10/01/2022 to 01/01/2023, inclusive, which consists of 62,895 suspicious liquidity pools and 22,195 tokens that are indicative of fraud patterns. We adopt a two-layer GAT as the backbone encoder. The GNN is pretrained for 1,000 epochs on the source graph with the hidden dimension fixed to 256 and 512. We apply edge dropping with rate 0.4 and feature masking with rate 0.1. Fine-tuning is performed for 500 epochs on the target graph. For each month in the transfer learning setting, we randomly split the dataset into training (10%), validation (10%), and testing (80%) subsets. The news texts are encoded using a frozen BERT-large-uncased model. The multi-modal branch uses a dropout rate of 0.1. All experiments are conducted on 4 NVIDIA RTX A5000 GPUs with 24GB memory. We use the Adam optimizer, and the learning rate and weight decays are tuned within the range $[1e^{-5}, 1e^{-1}]$. The scaling coefficients $\alpha = [\alpha_0, \alpha_1, \alpha_2, \alpha_3]$ are tuned within the range $[0.1, 10.0]$. The number of attention heads is set to 4. We compare our method with four state-of-the-art baselines: (i) GRACE_r [19], (ii) AdapterGNN [11], (iii) GTOT-Tuning [17], and (iv) GraphLoRA [15].

Findings Tables 1 and 2 show results of month-to-month transfer learning on Ethereum and Solana data. Overall, our MOE-GTL shows impressive results on both datasets. On Ethereum data (12 case studies), except 2018/02→2018/03 (but MOE-GTL is still the runner-up), MOE-GTL always outperforms all baselines with average relative gain of 1.12%. On Solana data, it is also clear that the proposed MOE-GTL achieves competitive results. Table 3 shows the performance of MOE-GTL by removing one of four components we leveraged (i.e., w/o FTE/NGE/MMFN/TMMD). We observe that all lead to consistent drops in performance, indicating that these components are complementary.

6 Conclusion

We have introduced the MOE-GTL framework that unifies transaction graph structures, fine-grained temporal features, and financial news via the multi-modal fusion network and temporal-aware maximum mean discrepancy. The MOE-GTL superior performance has been demonstrated via extensive experiments on two digital finance datasets. In future, we investigate whether the proposed framework can improve the graph-level anomaly detection in digital finance.

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